

DYNAMICAL SYSTEMS

PROBLEM SET 2

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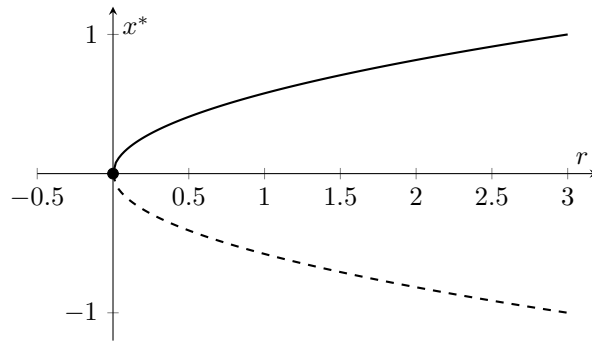
We begin by noting that all of the files referenced in this document are available at:

<https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet2>

The next exercises (3.4.5, 3.5.7, and 3.4.9) are designed to test your ability to distinguish among the various types of bifurcations — its easy to confuse them! In each case, find the values of r at which bifurcations occur, and classify those as: saddle-node, transcritical, supercritical pitchfork, or subcritical pitchfork. Finally sketch the bifurcation diagram of fixed points x^* vs r .

3.4.5 $\dot{x} = r - 3x^2$

Solution. Fixed points satisfy $x^* = \pm\sqrt{r/3}$ for $r \geq 0$. Since $f_x = -6x$, the positive branch is stable and the negative branch unstable. The two coalesce at $r = 0$, so this is a saddle-node bifurcation.



Solid is stable; dashed is unstable.

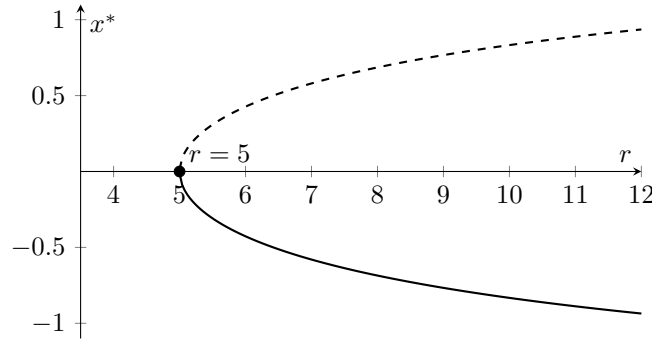
□

3.4.7 $\dot{x} = 5 - re^{-x^2}$

Solution. Fixed points require $e^{-x^2} = 5/r$. Thus none exist for $r < 5$, one ($x = 0$) exists at $r = 5$, and for $r > 5$,

$$x^* = \pm\sqrt{\ln(r/5)}.$$

Since $f_x = 2rxe^{-x^2} = 10x$ at equilibrium, the negative branch is stable and the positive branch unstable. Hence $r = 5$ is a saddle-node bifurcation.



The lower branch is stable and the upper branch is unstable.

□

3.4.9 $\dot{x} = x = \tanh(rx)$

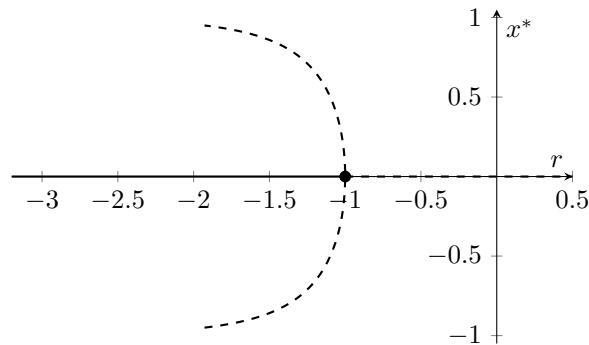
Solution. For $f = x + \tanh(rx)$, $x = 0$ is always fixed and $f_x(0) = 1 + r$, so the critical value is $r = -1$. Using

$$\tanh(rx) = rx - \frac{r^3 x^3}{3} + O(x^5),$$

we get near $r = -1$ the normal form $\dot{x} \approx \mu x + x^3/3$, where $\mu = r + 1$. Therefore the bifurcation at $r = -1$ is a subcritical pitchfork: the origin is stable for $r < -1$ and unstable for $r > -1$, with two unstable nonzero branches on the $r < -1$ side.

The nonzero fixed points may be plotted implicitly from

$$r = -\frac{\operatorname{arctanh}(x^*)}{x^*}.$$



Subcritical pitchfork at $r = -1$: solid is stable and dashed is unstable.

□

3.4.11 (An interesting bifurcation diagram) Consider $\dot{x} = rx - \sin(x)$

- For the case $r = 0$, find and classify all the fixed points, and sketch the vector field.
- Show that when $r > 1$, there is only one fixed point. What kind of fixed point is it?
- As r decreases from ∞ to 0, classify *all* the bifurcations that occur.
- For $0 < r \ll 1$, find an approximate formula for all values of r at which bifurcations occur.
- Now classify all the bifurcations that occur as r decreases from 0 to $-\infty$.
- Plot the bifurcation diagram for $-\infty < r < \infty$, and indicate the stability of the various fixed points.

Solution. Let $f = rx - \sin x$.

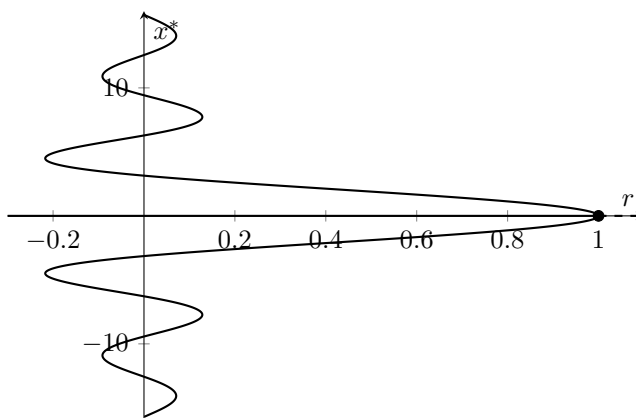
- (a) At $r = 0$, $x^* = n\pi$. Since $f_x = -\cos x$, $2k\pi$ is stable and $(2k+1)\pi$ unstable.
- (b) If $r > 1$, $|\sin x/x| \leq 1 < r$ for $x \neq 0$, so only $x = 0$ remains; it is unstable because $f_x(0) = r - 1 > 0$.
- (c) Saddle-nodes satisfy $f = f_x = 0$, hence $rx = \sin x$, $r = \cos x$, and therefore $\tan x = x$. At the origin, $f = (r-1)x + x^3/6 + \dots$, so $r = 1$ is a subcritical pitchfork.
- (d) For small positive r , roots of $\tan x = x$ lie near $a_n = (2n + \frac{1}{2})\pi$. Writing $x_n = a_n - \varepsilon_n$ gives $\varepsilon_n \sim 1/a_n$, hence

$$r_n \approx \frac{1}{(2n + \frac{1}{2})\pi}.$$

- (e) For $r < 0$, saddle-nodes again satisfy $\tan x = x$, now with $r = \cos x < 0$; these values accumulate at 0 from below.
- (f) The diagram is symmetric in x . The branch $x = 0$ is stable for $r < 1$ and unstable for $r > 1$; the other branches terminate in the saddle-nodes described above. Stability follows from the sign of $r - \cos x$.

The complete equilibrium curve is

$$r = \frac{\sin x^*}{x^*}, \quad r(0) = 1.$$



Stability along each segment is determined by the sign of $r - \cos x^*$.

□

3.4.14 (Subcritical pitchfork) Consider the system $\dot{x} = rx + x^3 - x^5$, which exhibits a subcritical pitchfork bifurcation.

- Find algebraic expressions for all the fixed points as r varies.
- Sketch the vector fields as r varies. Be sure to indicate all the fixed points and their stability.
- Calculate r_s , the parameter value at which the nonzero fixed points are born in a saddle-node bifurcation.

Solution. Write $\dot{x} = x(r + x^2 - x^4)$.

- (a) Besides $x = 0$, put $u = x^2$. Then $u^2 - u - r = 0$, so

$$x^* = \pm \sqrt{\frac{1 \pm \sqrt{1+4r}}{2}}.$$

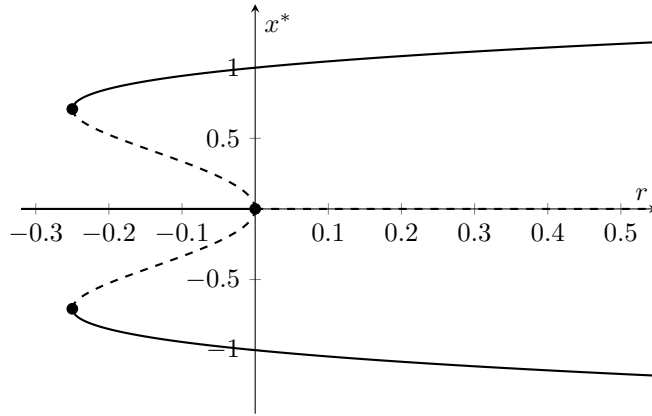
The outer pair exists for $r \geq -1/4$ and the inner pair for $-1/4 \leq r \leq 0$.

- (b) At $x = 0$, $f_x = r$. At a nonzero fixed point, $f_x = 2x^2(1 - 2x^2)$, so the inner pair is unstable and outer pair stable. Thus two saddle-nodes occur at $r = -1/4$, and the inner pair joins the origin in the subcritical pitchfork at $r = 0$.

- (c) The saddle-node value is

$$r_s = -\frac{1}{4},$$

with $x = \pm 1/\sqrt{2}$.



Solid branches are stable; dashed branches are unstable.

□

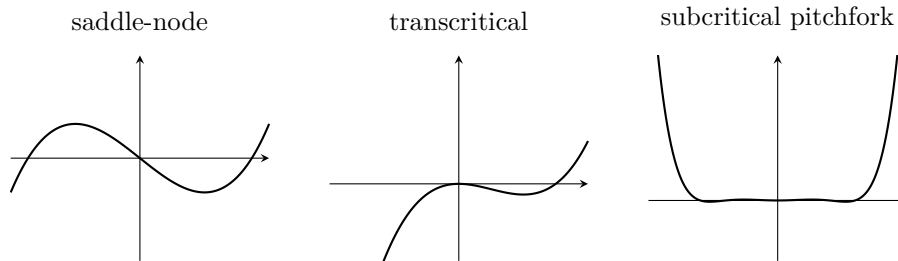
3.4.16 (Potentials) In parts (a)-(c) let $V(x)$ be the potential, in the sense that $\dot{x} = -dV/dx$. Sketch the potential as a function of r . Show all qualitatively different cases including bifurcation values of r .

- a) (Saddle-node) $\dot{x} = r - x^2$
 b) (Transcritical) $\dot{x} = rx - x^2$
 c) (Subcritical pitchfork) $\dot{x} = rx + x^3 - x^5$.

Solution. Since $\dot{x} = -V'(x)$:

- (a) $V = x^3/3 - rx + C$. For $r > 0$ it has a maximum at $-\sqrt{r}$ and minimum at $\sqrt{r}$; these merge at $r = 0$.
 (b) $V = -rx^2/2 + x^3/3 + C$. The stationary points 0 and r exchange maximum/minimum character at $r = 0$.
 (c) $V = -rx^2/2 - x^4/4 + x^6/6 + C$. The qualitative changes occur at $r = -1/4$ and $r = 0$: for $-1/4 < r < 0$ there are three minima separated by two maxima; for $r > 0$ the origin is a maximum and the two outer equilibria are minima.

Representative potential curves (additive constants are irrelevant) are:



□

3.5.4 (Bead on a horizontal wire) A bead of mass m is constrained to slide along a straight horizontal wire. A spring of relaxed length L_0 and spring constant k is attached to the mass and to a support point distance h from the wire given in Figure ?? below.

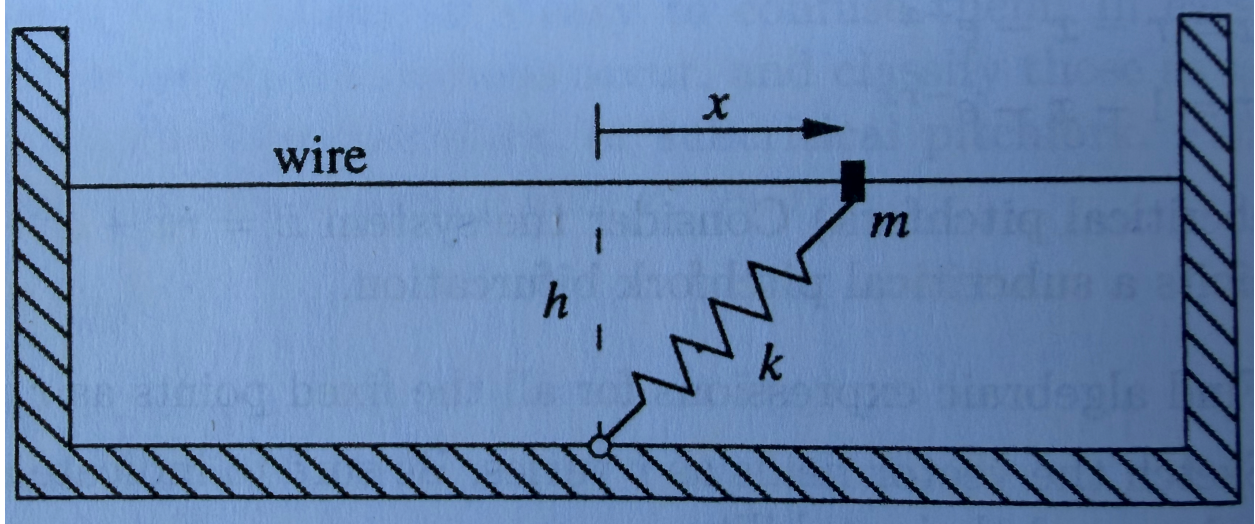


FIGURE 1.

Finally suppose that the motion of the bead is opposed by a viscous damping force $b\dot{x}$.

- Write Newton's law for the motion of the fixed bead.
- Find all possible equilibria, i.e. fixed points, as functions of k , h , m , b , and L_0 .
- Suppose $m = 0$. Classify the stability of all the fixed points, and draw a bifurcation diagram.
- If $m \neq 0$, how small must m be, to be considered negligible? In what sense is it negligible?

Solution. Let $\ell = \sqrt{x^2 + h^2}$.

- (a) Newton's law is

$$m\ddot{x} + b\dot{x} + kx \left(1 - \frac{L_0}{\sqrt{x^2 + h^2}}\right) = 0.$$

- (b) Equilibria satisfy $x(1 - L_0/\ell) = 0$, so $x = 0$ always and, if $L_0 \geq h$,

$$x = \pm \sqrt{L_0^2 - h^2}.$$

- (c) If $m = 0$, $f'(0) = -(k/b)(1 - L_0/h)$. Thus the origin is stable for $L_0 < h$ and unstable for $L_0 > h$, while the two nonzero equilibria are stable. Hence $L_0 = h$ is a supercritical pitchfork.
- (d) Inertia is negligible when its relaxation time m/b is much shorter than the overdamped time b/k , i.e.

$$\frac{mk}{b^2} \ll 1.$$

Then inertia matters only during a short initial transient.

□

3.6.2 (Imperfect transcritical bifurcation) Consider the system $\dot{x} = h + rx - x^2$. When $h = 0$, this system undergoes a transcritical bifurcation at $r = 0$. Our goal is to see how the bifurcation diagram of x^* vs. r is affected by the imperfection parameter h .

- Plot the bifurcation diagram for $\dot{x} = h + rx - x^2$, for $h < 0$, $h = 0$, and $h > 0$.

- b) Sketch the regions in the (r, h) plane that correspond to qualitatively different vector fields, and identify the bifurcation that occur on the boundaries of those regions.
- c) Plot the potential $V(x)$ corresponding to all the different regions in the (r, h) plane.

Solution. The fixed points are

$$x_{\pm}^* = \frac{r \pm \sqrt{r^2 + 4h}}{2}.$$

- (a) For $h > 0$ there are always two fixed points, with x_+ stable and x_- unstable. For $h = 0$, $x = 0$ and $x = r$ exchange stability at the transcritical bifurcation $r = 0$. For $h < 0$, saddle-nodes occur at $r = \pm 2\sqrt{-h}$; there are no fixed points between them.
- (b) The bifurcation set in parameter space is

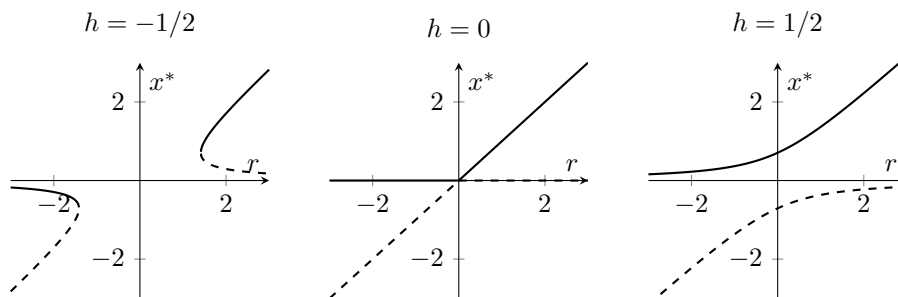
$$h = -\frac{r^2}{4}.$$

Above it are two fixed points, on it one double fixed point, and below it none. Away from the origin the boundary consists of saddle-node bifurcations.

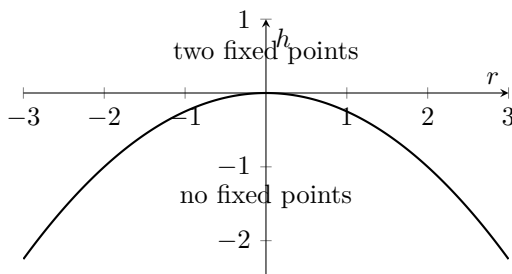
- (c) The potential is

$$V(x) = -hx - \frac{r}{2}x^2 + \frac{1}{3}x^3 + C.$$

Two stationary points occur when $r^2 + 4h > 0$, one degenerate stationary point on the parabola, and none below it.



Solid branches are stable; dashed branches are unstable.



□

3.7.5 (A biochemical switch) Zebra stripes and butterfly wing patterns are two of the most spectacular examples of biological pattern formation. Explaining the development of these patterns is one of the outstanding problems of Biology; see Murray (2003) for an excellent review.

As one ingredient in a model of pattern formation, Lewis et al. (1977) considered a simple example of a biochemical switch, in which a gene G is activated by a biochemical signal substance S . For example, the gene may normally be inactive but can be “switched on” to produce a pigment or other gene product when

concentration of S exceeds a certain threshold. Let $g(t)$ denote the concentration of the gene product, and assume that the concentration s_0 of S is fixed. The model is

$$\dot{g} = k_1 s_0 - k_2 g + \frac{k_3 g^2}{k_4 + g^2},$$

where k 's are positive constants. The production of g is stimulated by s_0 at rate k_1 , and by an *autocatalytic* (positive feedback) process, modeled by the non-linear term. There is also a linear degradation of g at a rate k_2 .

- a) Show that the system can be put in the dimensionless form

$$\frac{dx}{d\tau} = s - rx + \frac{x^2}{1+x^2},$$

where $r > 0$ and $s \geq 0$ are dimensionless groups.

- b) Show that if $s = 0$, there are two positive fixed points x^* if $r < r_c$, where r_c is to be determined.
 c) Assume that initially there is no gene product, i.e. $g(0) = 0$, and suppose s is slowly increased from 0 (the activating signal is turned on); what happens to $g(t)$? What happens if s then goes back to zero? Does the gene turn off again?
 d) Find parametric equations for the bifurcation curves in (r, s) space, and classify the bifurcations that occur.
 e) Compute a quantitatively accurate plot of the stability diagram in (r, s) space.

For further discussion of this model, see Lewis et al. (1977); Edelstein-Kishet (1988); Section 7.5; or Murray (2002), Chapter 6.

Solution. (a) Let $x = g/k_4$ and $\tau = (k_3/k_4)t$. Then

$$\frac{dx}{d\tau} = s - rx + \frac{x^2}{1+x^2}, \quad s = \frac{k_1 s_0}{k_3}, \quad r = \frac{k_2 k_4}{k_3}.$$

- (b) For $s = 0$, positive fixed points satisfy $r = x/(1+x^2)$. This has maximum $1/2$ at $x = 1$, so $r_c = 1/2$ and two positive fixed points exist for $0 < r < 1/2$.
 (c) Slowly increasing s follows the low stable branch until a saddle-node causes a jump to the high branch. On decreasing s again, the system can remain on the high branch: the switch exhibits hysteresis.
 (d) From $f = f_x = 0$,

$$r(x) = \frac{2x}{(1+x^2)^2}, \quad s(x) = \frac{x^2(1-x^2)}{(1+x^2)^2}.$$

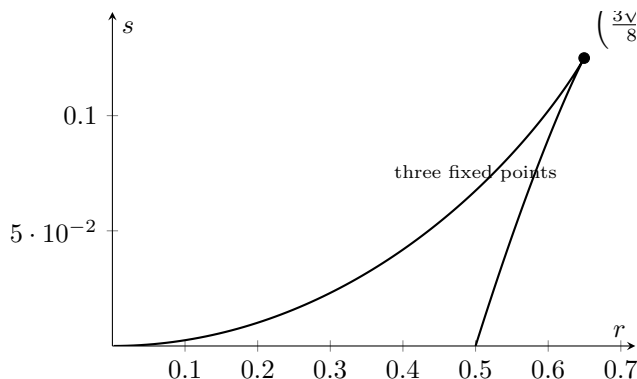
These parameterize the saddle-node curves.

- (e) The cusp satisfies $dr/dx = 0$, giving $x = 1/\sqrt{3}$ and

$$(r, s) = \left(\frac{3\sqrt{3}}{8}, \frac{1}{8} \right).$$

Inside the cusp region there are three fixed points (two stable, one unstable); outside there is one.

The saddle-node boundary is given parametrically by the formulas above:



□

5.1.10 (Attracting and Liapunov stable) here are the official definitions of various types of stability. Consider a fixed point $\mathbf{x}^*$ of a system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$.

We say that $\mathbf{x}^*$ is **attracting** if there is a $\delta > 0$ such that $\lim_{t \rightarrow \infty} x(t) = \mathbf{x}^*$ whenever $\|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$ within a distance of δ to $\mathbf{x}^*$ is guaranteed to converge to $\mathbf{x}^*$ *eventually*. As shown schematically in Figure ??, trajectories that start nearby are allowed to stray from $\mathbf{x}^*$ in the short run, but they must approach $\mathbf{x}^*$ in the long run.

In contrast, Liapunov stability requires that nearby trajectories remain close for *all* time. We say that $\mathbf{x}^*$ is **Liapunov stable** if for each $\varepsilon > 0$ there is a $\delta > 0$ such that $\|\mathbf{x}(t) - \mathbf{x}^*\| < \varepsilon$ whenever $t \geq 0$ and $\|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$. Thus, trajectories that start within δ of $\mathbf{x}^*$ remain within ε of $\mathbf{x}^*$ for all positive time. Figure ?? gives a good example of this. Finally $\mathbf{x}^*$ is **asymptotically stable** if it is both attracting and

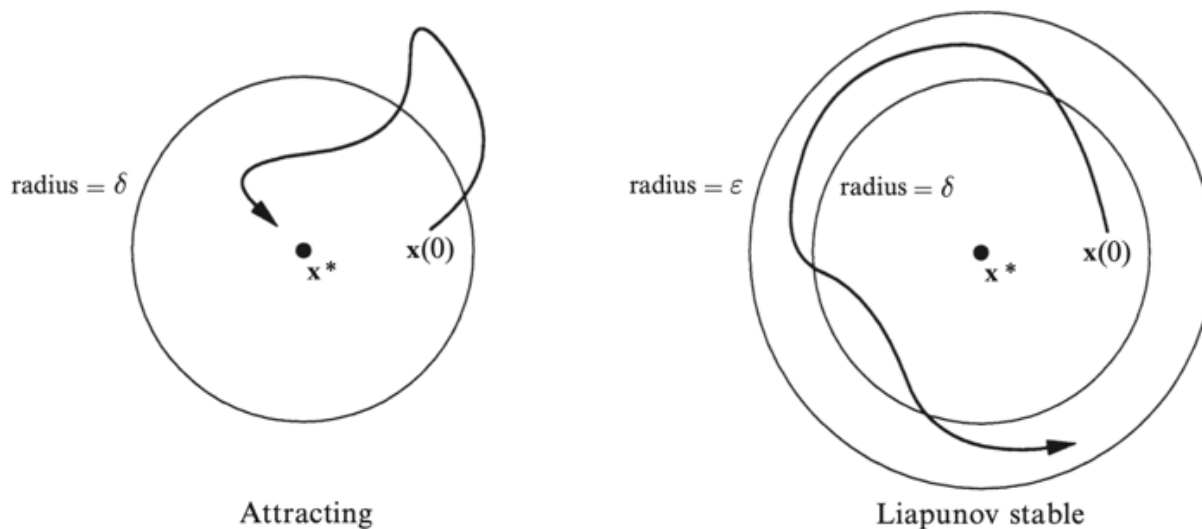


FIGURE 2.

Liapunov stable.

For each of the following systems, decide whether the origin is attracting, Liapunov stable, asymptotically stable, or none of the above.

- a) $\dot{x} = y, \dot{y} = -4x$
- b) $\dot{x} = 2y, \dot{y} = x$

- c) $\dot{x} = 0, \dot{y} = x$
- d) $\dot{x} = 0, \dot{y} = -y$
- e) $\dot{x} = -x, \dot{y} = -5y$
- f) $\dot{x} = x, \dot{y} = y$

Solution. (a) Eigenvalues $\pm 2i$: closed ellipses. The origin is Liapunov stable but not attracting.
 (b) Eigenvalues $\pm\sqrt{2}$: saddle. The origin is neither attracting nor Liapunov stable.
 (c) $x = x_0, y = y_0 + x_0 t$: the origin is neither attracting nor Liapunov stable.
 (d) $x = x_0, y = y_0 e^{-t}$: Liapunov stable but not attracting.
 (e) $x = x_0 e^{-t}, y = y_0 e^{-5t}$: asymptotically stable, hence attracting and Liapunov stable.
 (f) $x = x_0 e^t, y = y_0 e^t$: neither attracting nor Liapunov stable.

□

5.2.1 Consider the system $\dot{x} = 4x - y, \dot{y} = 2x + y$.

- a) Write the system as $\dot{\mathbf{x}} = A\mathbf{x}$. Show that the characteristic equation is $\lambda^2 - 5\lambda + 6 = 0$, and find the eigenvalues and eigenvectors of A .
- b) Find the general solution of the system.
- c) Classify the fixed point at the origin.
- d) Solve the system subject to the initial condition $(x_0, y_0) = (3, 4)$.

Solution. (a)

$$A = \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix}, \quad \det(A - \lambda I) = \lambda^2 - 5\lambda + 6.$$

Thus $\lambda = 2, 3$, with eigenvectors $(1, 2)^T$ and $(1, 1)^T$, respectively.

(b)

$$\mathbf{x}(t) = c_1 e^{2t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

- (c) Both eigenvalues are positive, so the origin is an unstable node.
- (d) From $(3, 4)$, $c_1 = 1, c_2 = 2$. Hence

$$x(t) = e^{2t} + 2e^{3t}, \quad y(t) = 2e^{2t} + 2e^{3t}.$$

□

5.2.6 Plot the phase portrait and classify the fixed point of the following linear systems. If the eigenvectors are real, indicate them in your sketch.

$$\begin{aligned} \dot{x} &= -3x + 2y \\ \dot{y} &= x - 2y \end{aligned}$$

Solution. Here

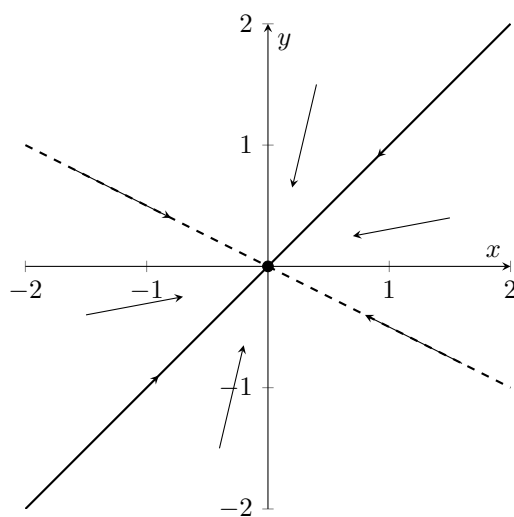
$$A = \begin{pmatrix} -3 & 2 \\ 1 & -2 \end{pmatrix},$$

and

$$\det(A - \lambda I) = \lambda^2 + 5\lambda + 4 = (\lambda + 1)(\lambda + 4).$$

Thus $\lambda_1 = -1$ with eigenvector $(1, 1)^T$, and $\lambda_2 = -4$ with eigenvector $(-2, 1)^T$. Both eigenvalues are real, distinct, and negative, so the origin is a stable node. The phase portrait has invariant straight lines along

$y = x$ and $y = -x/2$, and all trajectories approach the origin as $t \rightarrow \infty$, asymptotically tangent to the slower eigendirection $y = x$.



The solid line is the slow eigendirection $y = x$ and the dashed line is $y = -x/2$. All trajectories approach the origin.

□